Benchmarking GPT-4o for TNM Staging in Colorectal Cancer using a Feature Extraction-Based Method

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BACKGROUND

- **Colorectal cancer (CRC)** is a leading cause of cancer-related mortality worldwide.
- **TNM staging** is a cornerstone of CRC management.
- Manual staging is resource-intensive and prone to variability.
- Advances in **natural language processing (NLP)** have enabled the development of models capable of interpreting complex medical texts.
- **GPT-4o**, a large language model (LLM), exhibits enhanced logical reasoning capabilities.
- **The performance of GPT-4o in CRC TNM** staging has not been thoroughly evaluated.
- This study aims to **assess GPT-4o against a pathology** in accurately assigning TNM stages to CRC cases.

HYPOTHESIS

GPT-4o will demonstrate superior performance in TNM staging compared to the pathologist.

METHODS

- Obtained **100 pathology reports** from **The Cancer Genome Atlas** (TCGA) Colon Adenocarcinoma **(COAD) dataset**
- We used an **extraction-based approach** in initial prompt to identify:
  1) Tumour extension
  2) Lymph node involvement
  3) Distant metastasis
- GPT-4o was prompted again with these features
- **8th edition AJCC guidelines** used for TNM staging criteria
- Classifications compared to **ground truth determined by a pathologist**

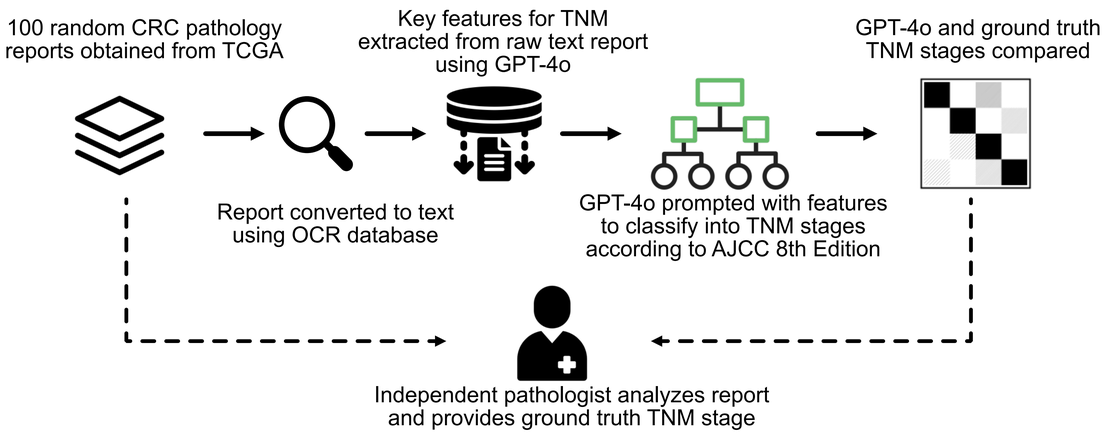

Figure 1. Overview of pipeline methodology

RESULTS

STAGE CLASSIFICATION	N
T-stage	
Tis	1
T1	2
T2	10
T3	67
T4a	7
T4b	7
N-stage	
NX	2
N0	60
N1	1
N1a	8
N1b	11
N2a	6
N2b	5
M-stage	
MX	86
M0	1
M1a	4
M1b	1
M1c	1

Table 1. Overview of **TCGA CRC cohort** of the 100 cases chosen for this study.

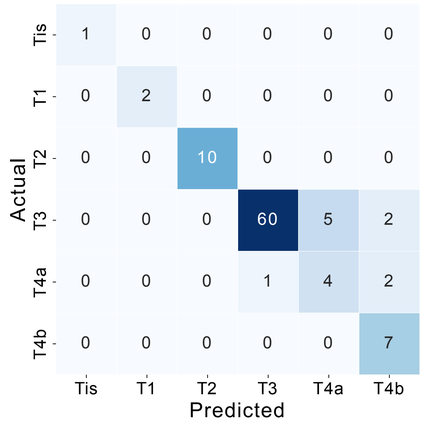

Figure 2. Confusion matrix of **GPT-4o vs ground-truth** in predicting **T-stage**. Average precision = 0.84, F1 score = 0.86, accuracy = 0.89

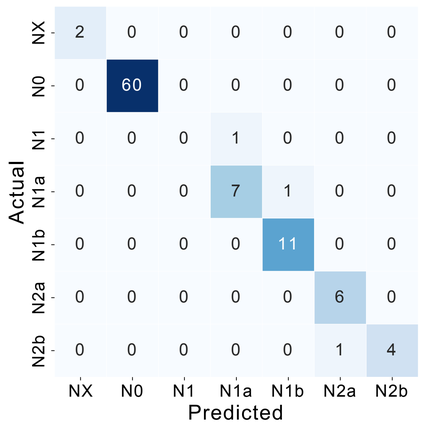

Figure 3. Confusion matrix of **GPT-4o vs ground-truth** in predicting **N-stage**. Average precision = 0.84, F1 score = 0.81, accuracy = 0.96

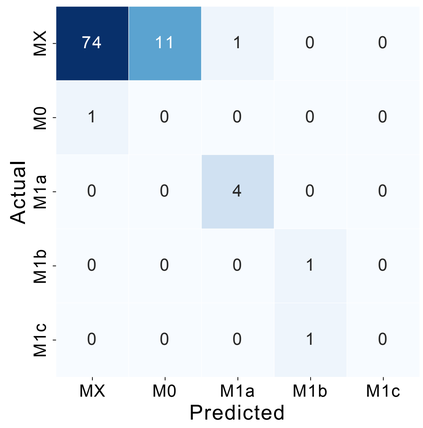

Figure 4. Confusion matrix of **GPT-4o vs ground-truth** in predicting **M-stage**. Average precision = 0.46, F1 score = 0.49, accuracy = 0.85

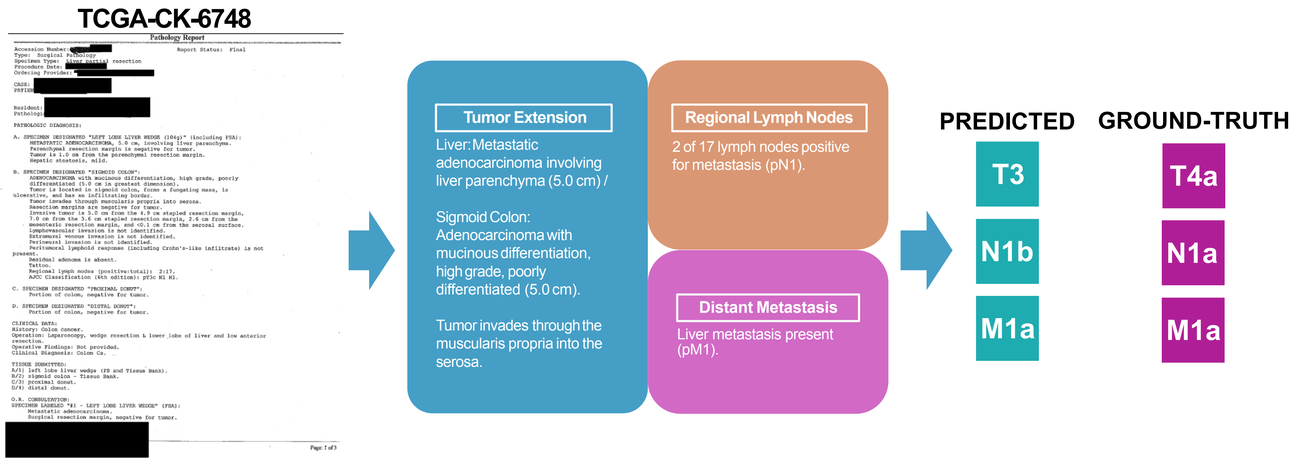

Figure 5. One example of a pathology report that conflicted with ground-truth pathologist's interpretation.

DISCUSSION

- **Novel AI comparison**: This study presents a new evaluation of AI models for **TNM staging in CRC**, highlighting case-specific strengths and limitations of different NLP approaches.
- **Open-source LLM advantages**: Findings reinforce evidence that **reasoning-based LLMs** can enhance automation of clinical tasks involving unstructured data.
- **Future directions**: Research should explore **real-world integration** into pathology workflows and **scalability across other cancer types**